Easy to plot

```
> def plot_lines(x_values, line_type, amplitude, mean, gamma, sigma,y0):···

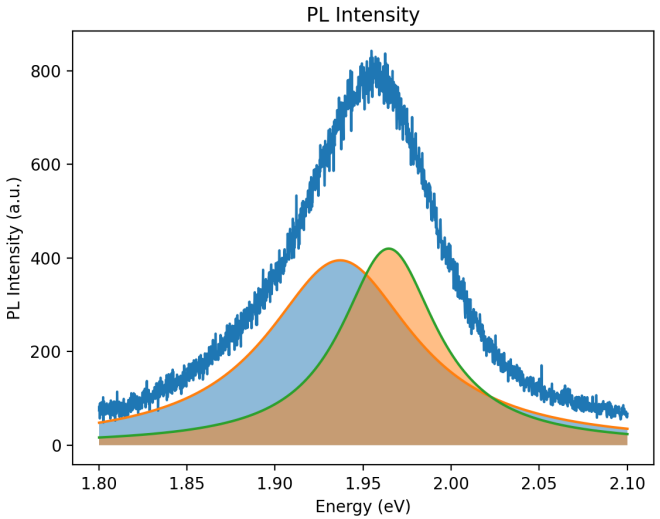
  Tabnine: Test | Explain | Document | Ask
> def plot_data(kind,allornot,savefig,shadow,line_color='blue',shadow_color='gray'):···
  # Before using, rename the txt file into "xx_xx_number", the number will be a symbol to sort the plots
  # 'PL','Raman','g2'
  #  single means draw each txt file with a single plot; all means draw all files with a summary plot
  plot_lines(x_values,'lorentz',395,1.93702,0.10177,0,0)
  plot_lines(x_values,'lorentz',420,1.96459,0.06629,0,0)
  plot_data( 'PL', 'all', savefig=True, shadow=False, line_color='red', shadow_color='green')
```

ALL plots.py M ● | box1_PL_01.txt U ×

box1_PL_01.txt > data

	#Wave	#Intensity
1		
2	1.800184	60.506870
3	1.800320	69.150711
4	1.800456	62.656475
5	1.800592	43.203529
6	1.800728	75.592476
7	1.800864	51.825447
8	1.801000	51.816051
9	1.801136	62.599716
10	1.801272	51.797272
11	1.801408	64.734848

Rename, Choose and Plot

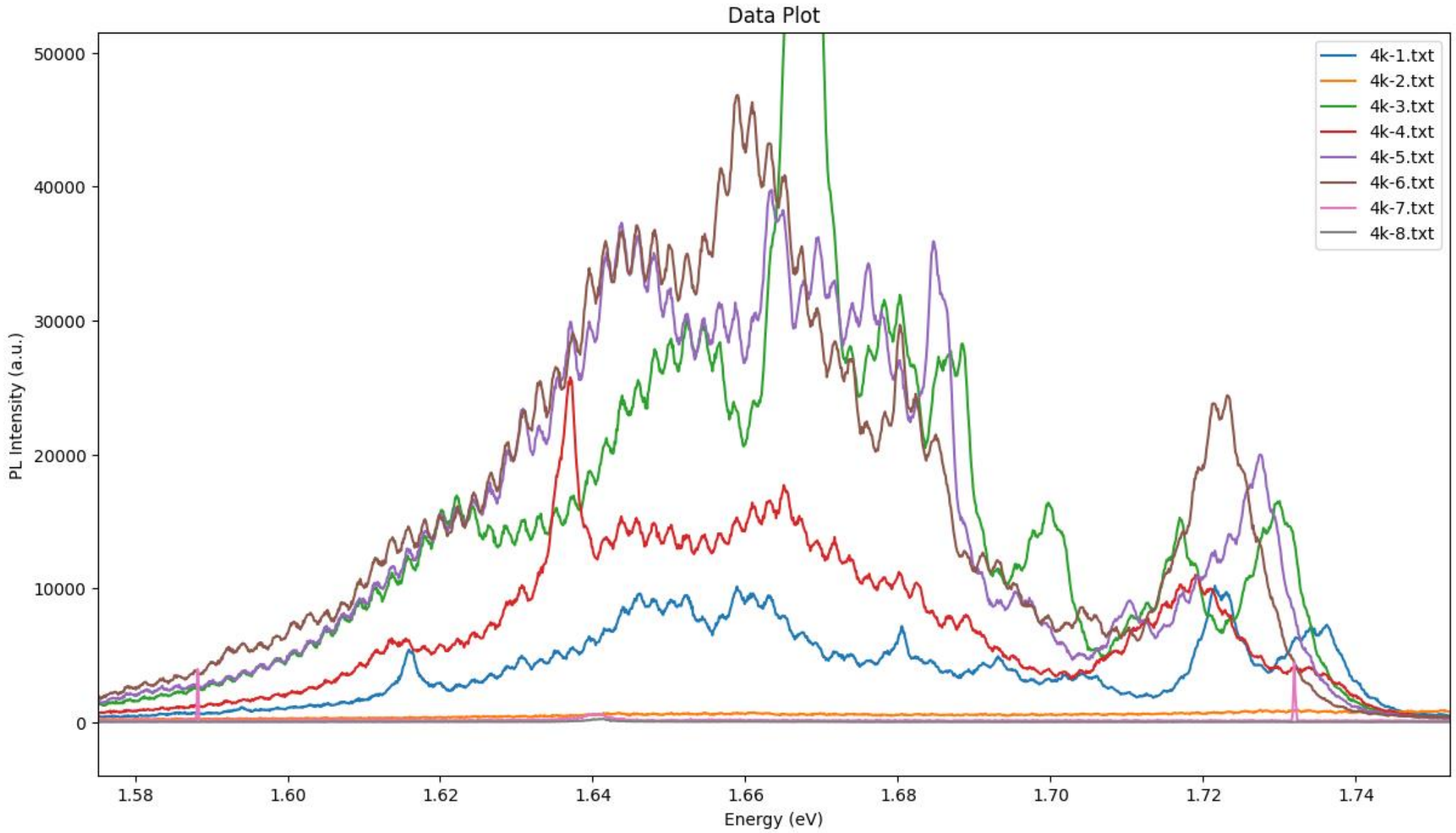


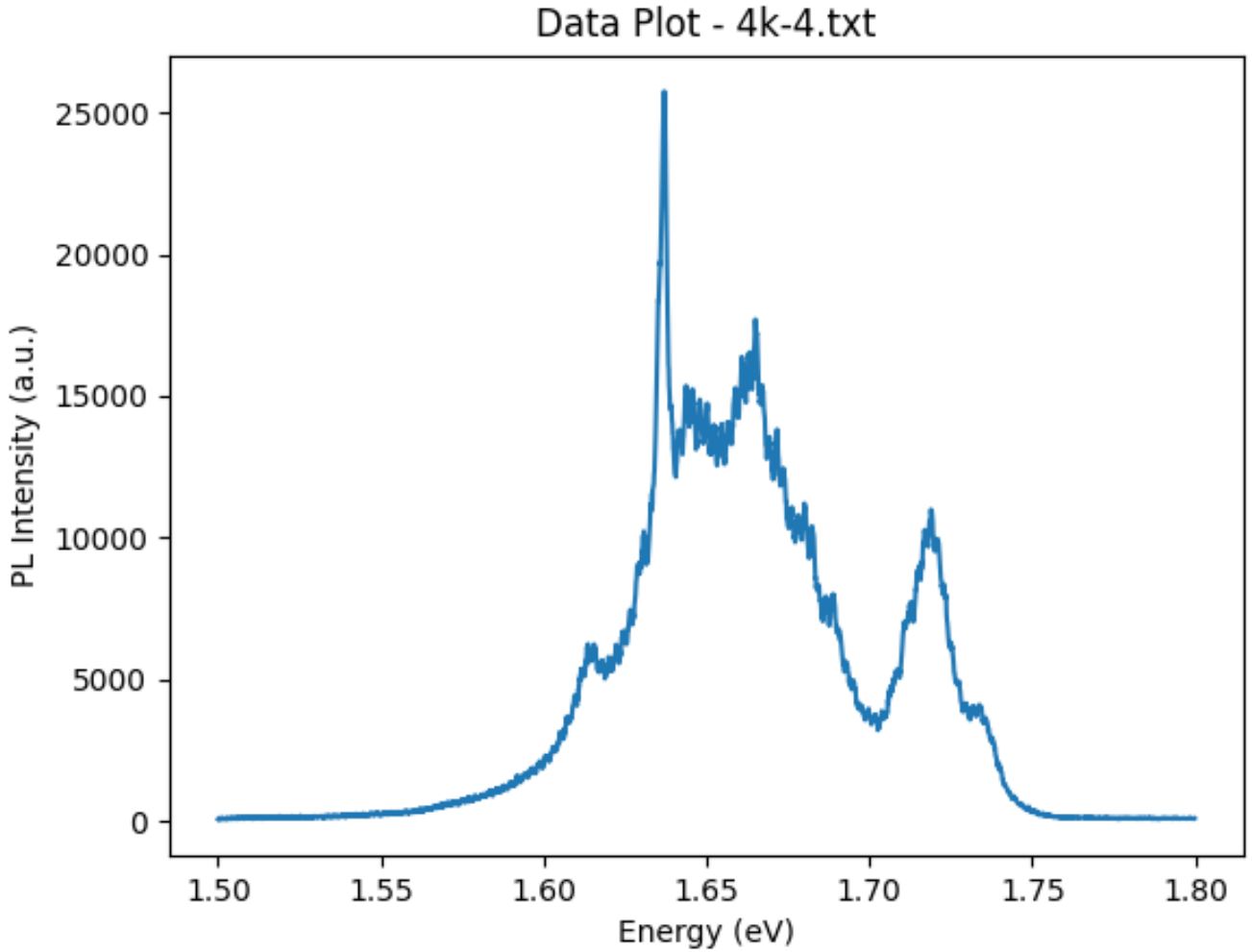
名称

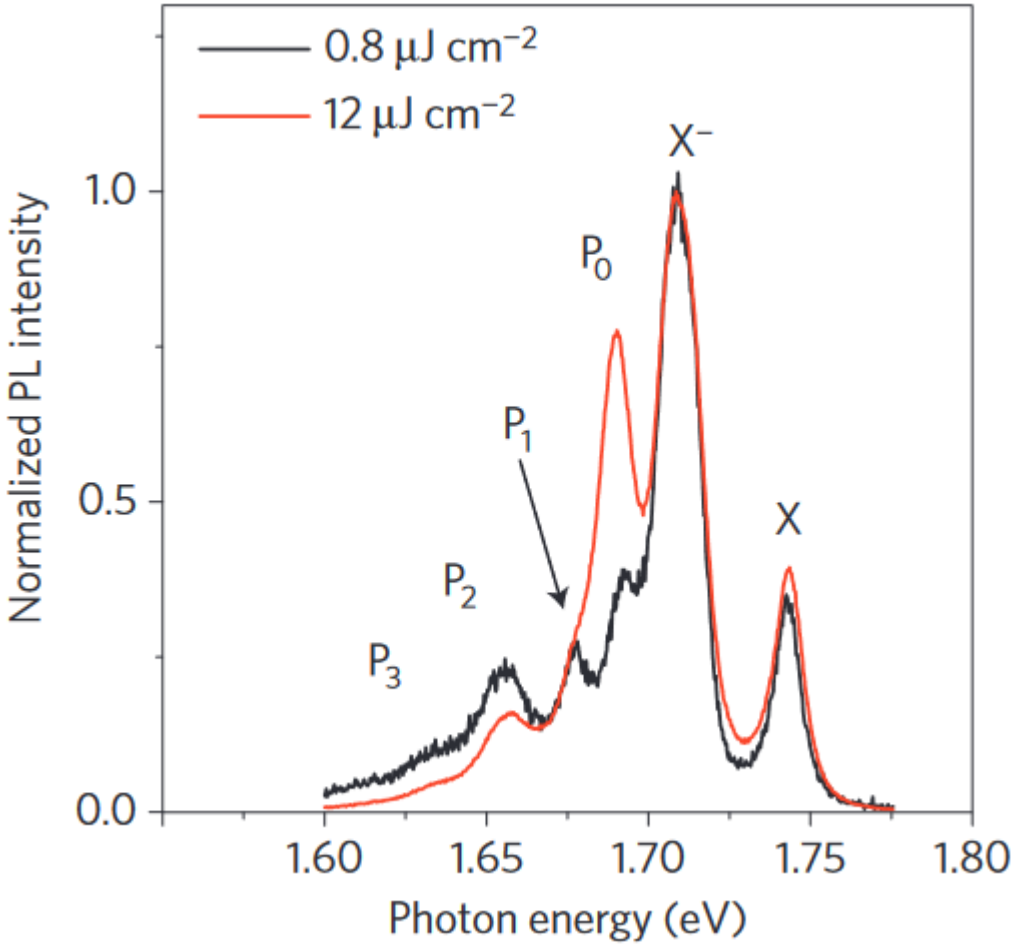
- PL_all_data_plot
- ALL plots
- box1_PL_01
- Bilayer
- monolayer
- .vscode
- .venv

Monolayer WSe_2

WSe2 Monolayer PL

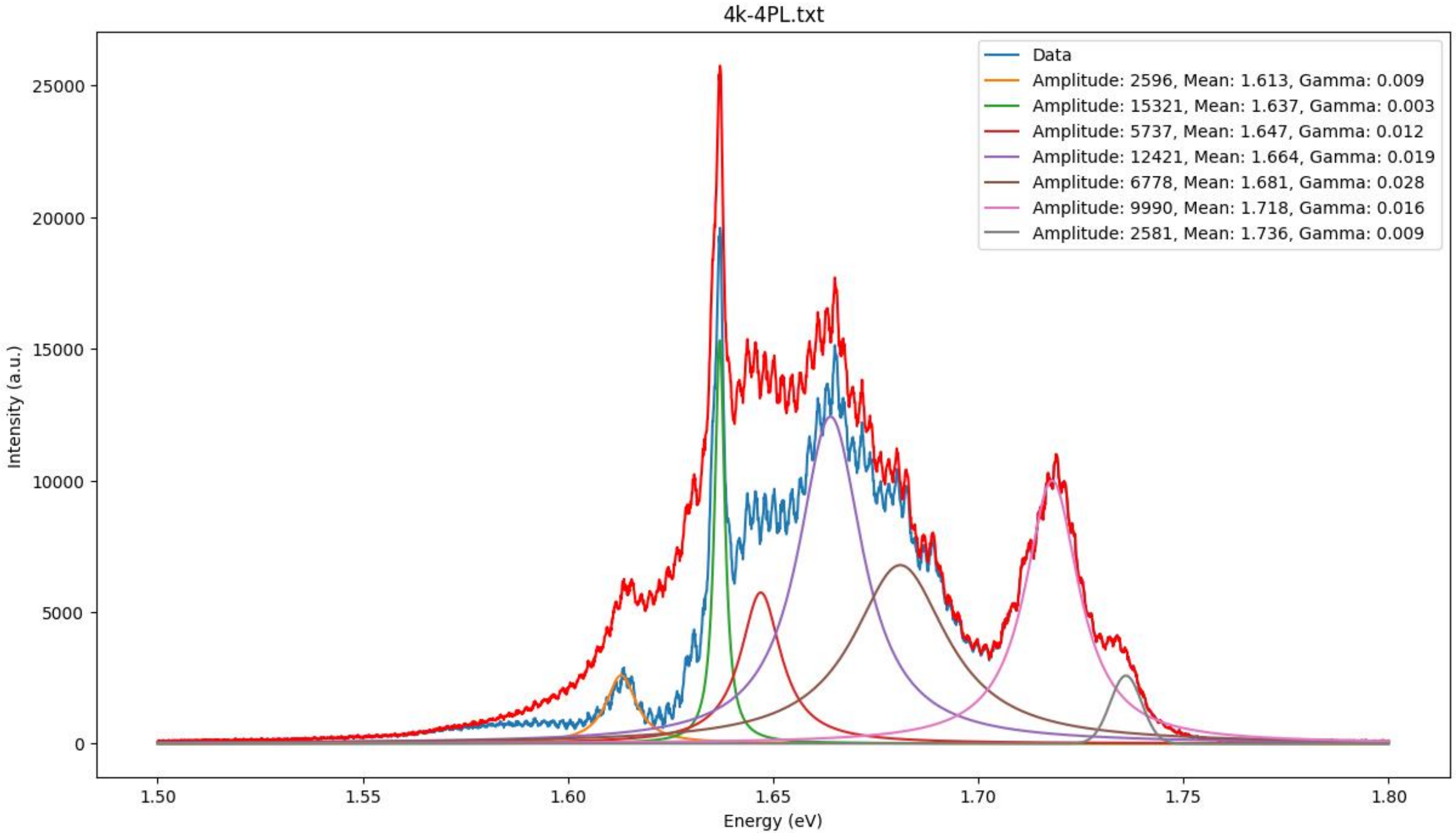




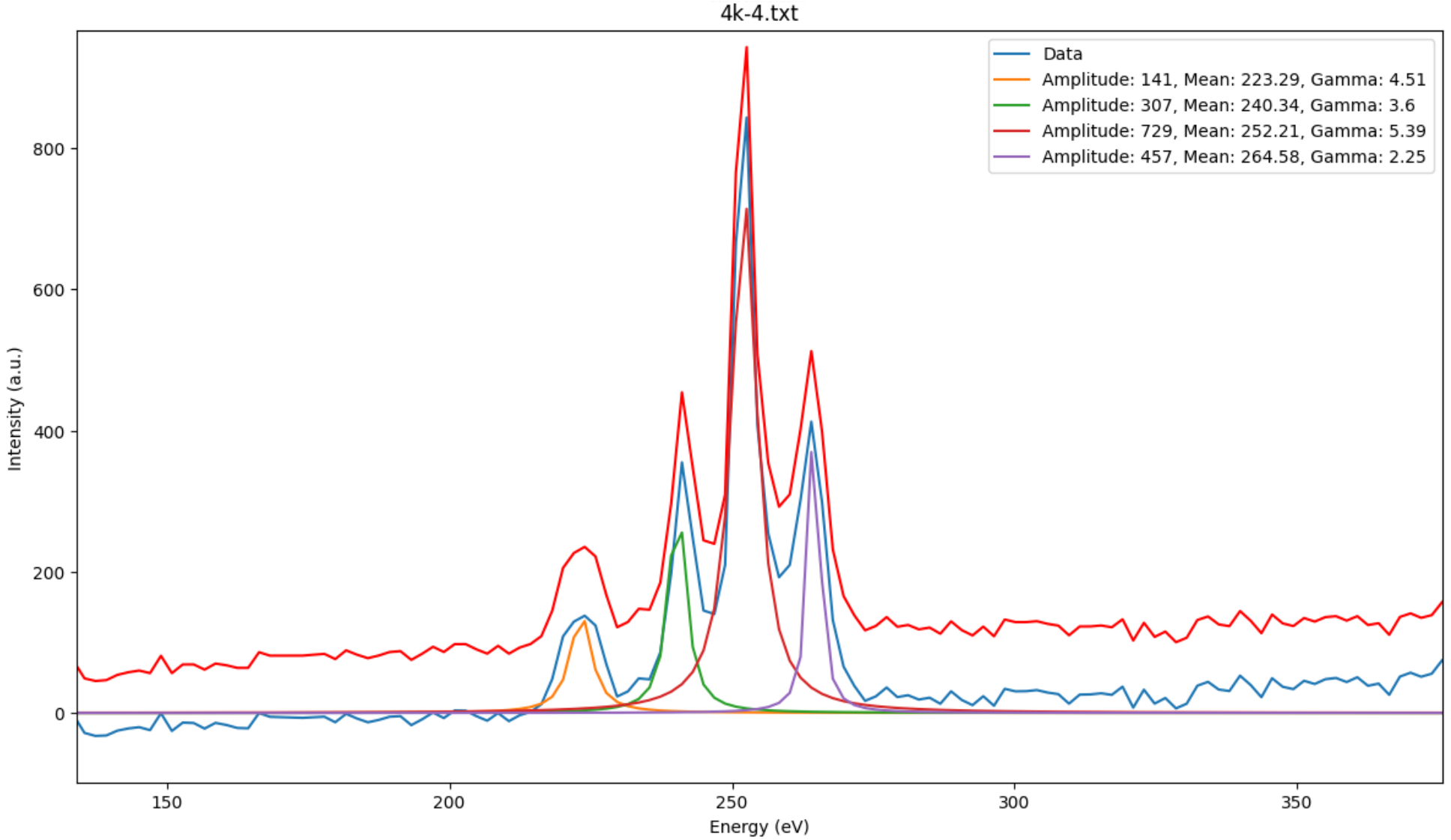


1.7-1.75eV peaks: radiative recombination of neutral excitons (1.73)/negative trions (1.71)
1.65-1.70eV peaks: defect-related localized state transition (1.66) biexciton emission (1.68)
1.63eV main peak: defect-related localized state transition
1.61eV peak: recombination of A exciton (lowest-energy exciton transition)

DOI: 10.1038/NPHYS3324 DOI: 10.1038/srep22414


4k-4PL.txt
Data
Amplitude: 2596, Mean: 1.613, Gamma: 0.009
Amplitude: 15321, Mean: 1.637, Gamma: 0.003
Amplitude: 5737, Mean: 1.647, Gamma: 0.012
Amplitude: 12421, Mean: 1.664, Gamma: 0.019
Amplitude: 6778, Mean: 1.681, Gamma: 0.028
Amplitude: 9990, Mean: 1.718, Gamma: 0.016
Amplitude: 2581, Mean: 1.736, Gamma: 0.009
Intensity (a.u.)
25000
20000
15000
10000
5000
0
1.50
1.55
1.60
1.65
1.70
1.75
1.80
Energy (eV)

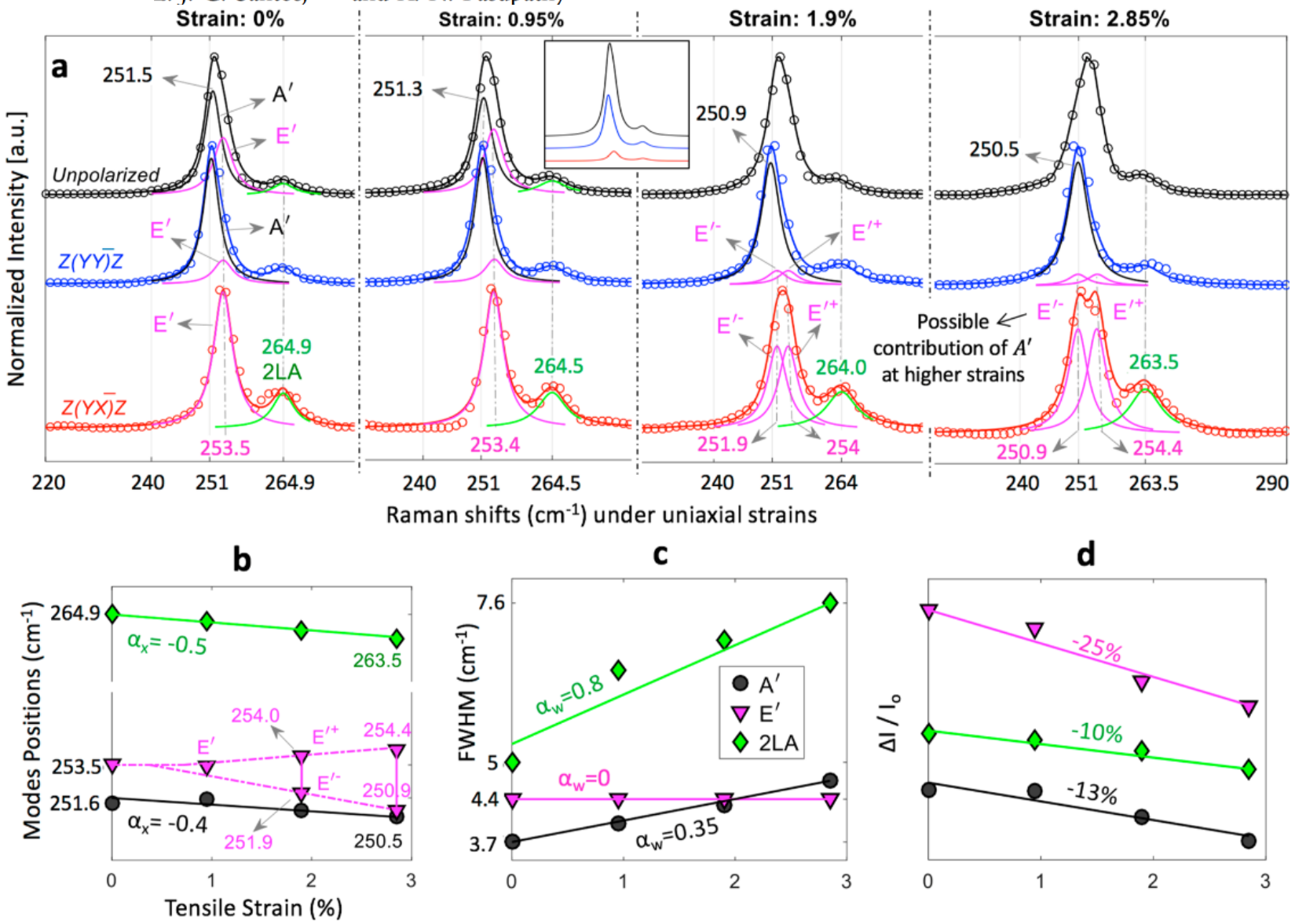
Monolayer Raman



223.3cm^{-1}: E'^{-} 240.3cm^{-1}: A' 252.2cm^{-1}: E'^{+} 264.6cm^{-1}: $2LA$

Strain Engineering and Raman Spectroscopy of Monolayer Transition Metal Dichalcogenides

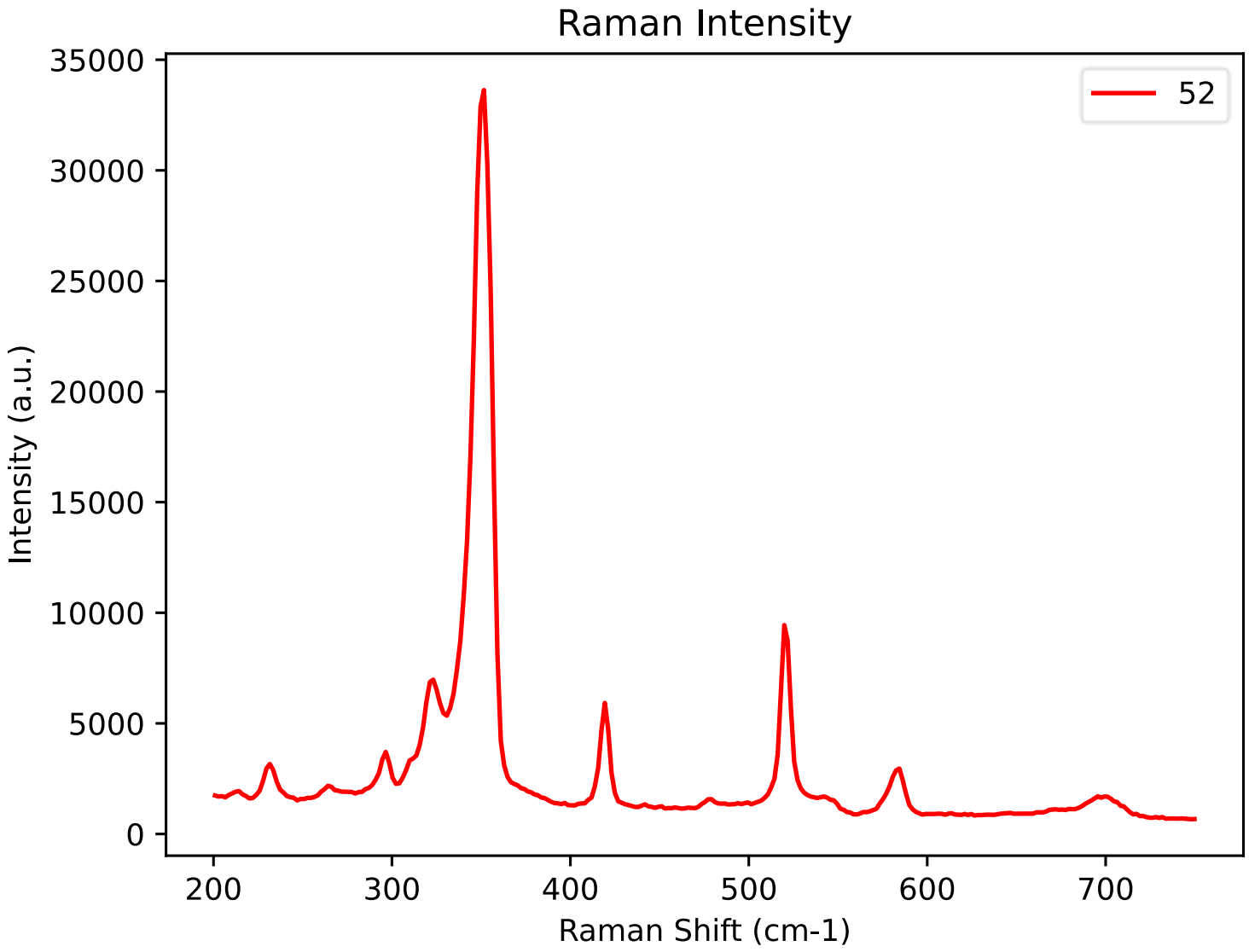
A. M. Dadgar,[†,‡] D. Scullion,[§] K. Kang,[∥] D. Esposito,[⊥] E. H. Yang,[∥] I. P. Herman,[#] M. A. Pimenta,[▽] E.-J. G. Santos,[*,○] and A. N. Pasupathy[*,‡]

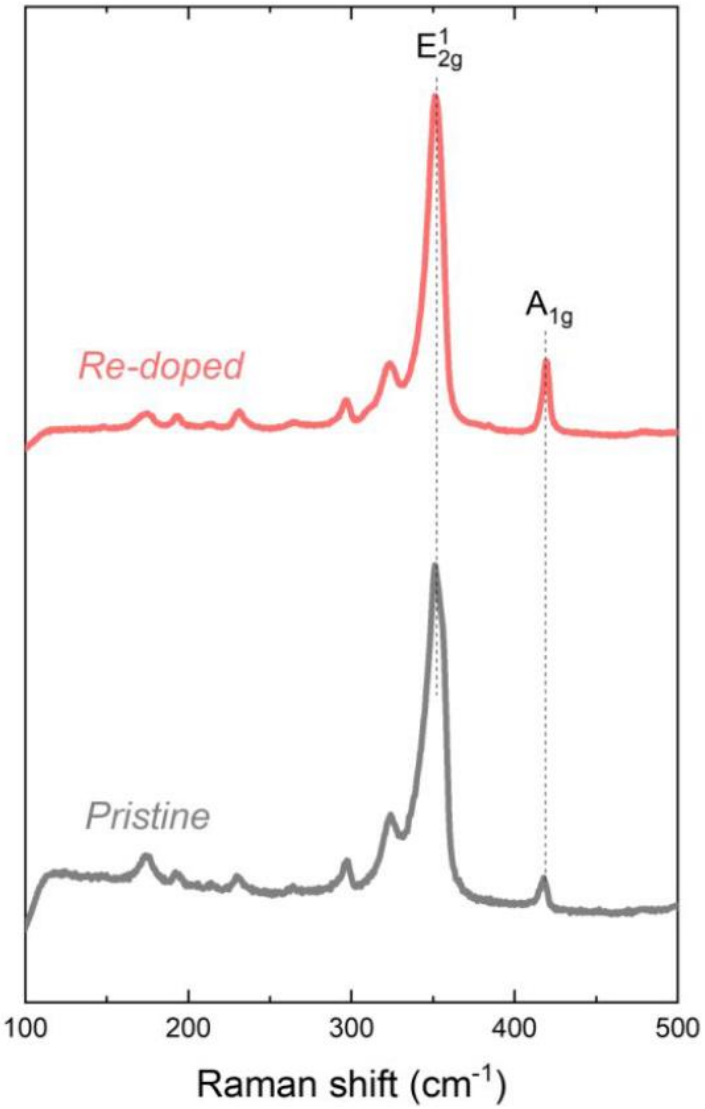


Chem. Mater. 2018, 30, 5148−5155

Re-doped WS2

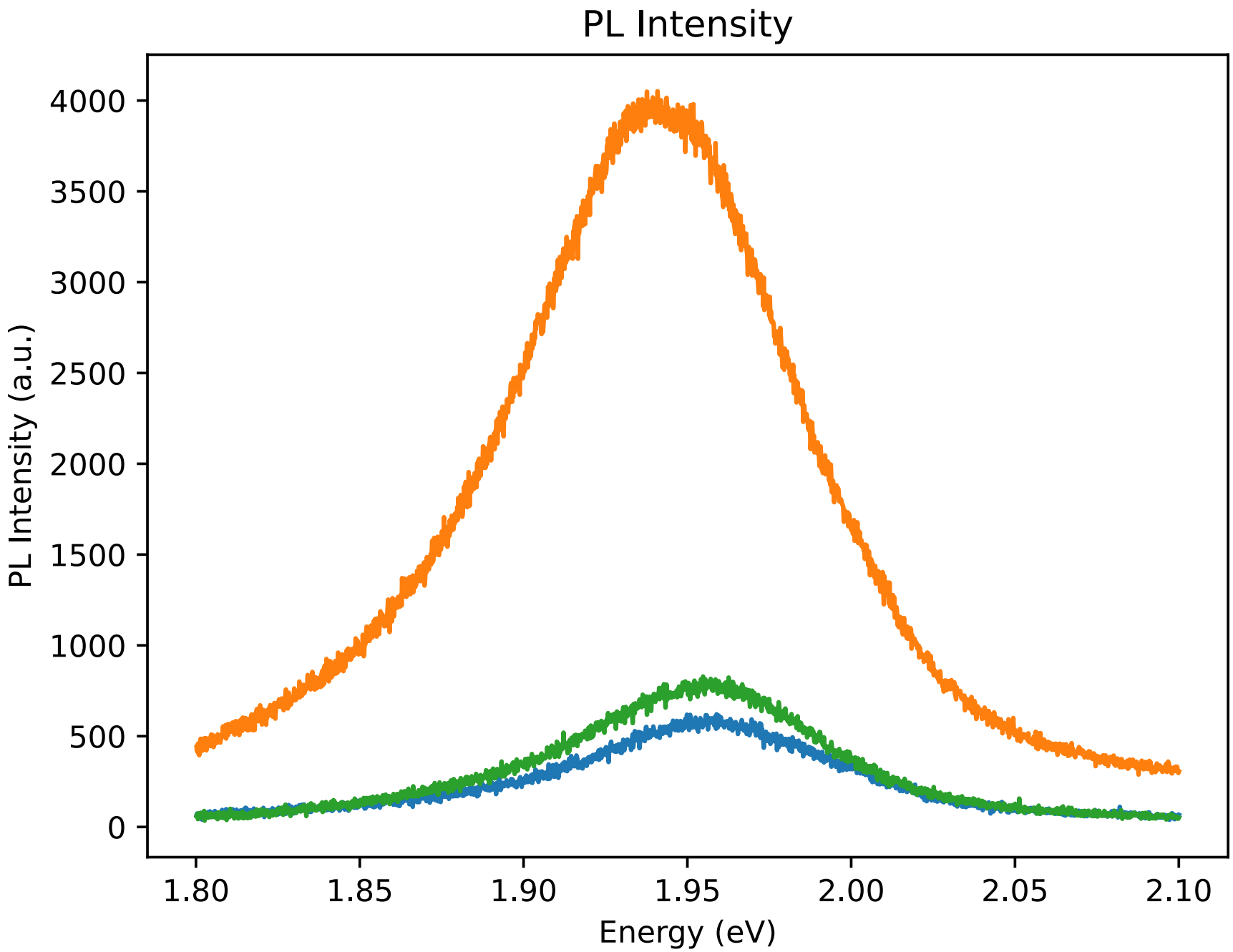
Re-WS2 Raman

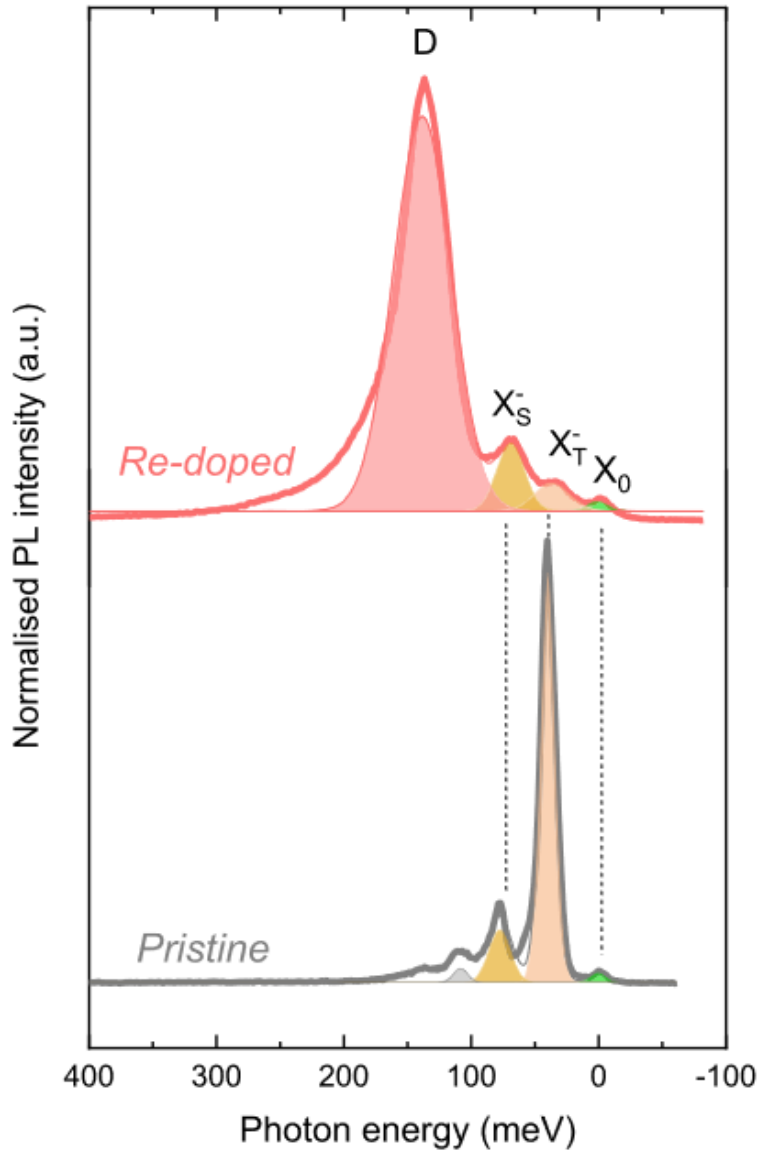




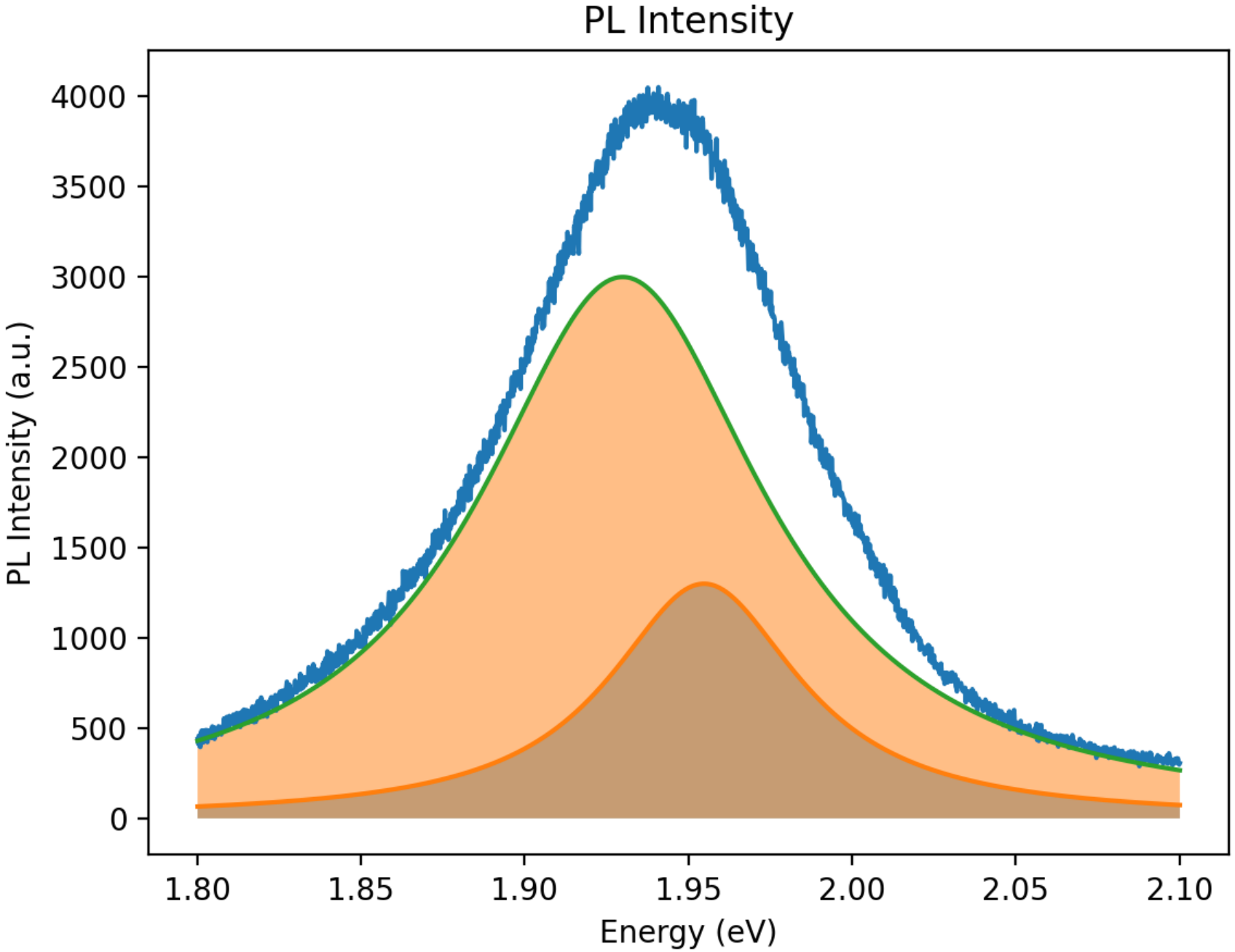
Nano Lett. 2021, 21, 5293−5300

Re-WS2 PL (RT)


PL Intensity
PL Intensity (a.u.)
4000
3500
3000
2500
2000
1500
1000
500
0
1.80
1.85
1.90
1.95
2.00
2.05
2.10
Energy (eV)

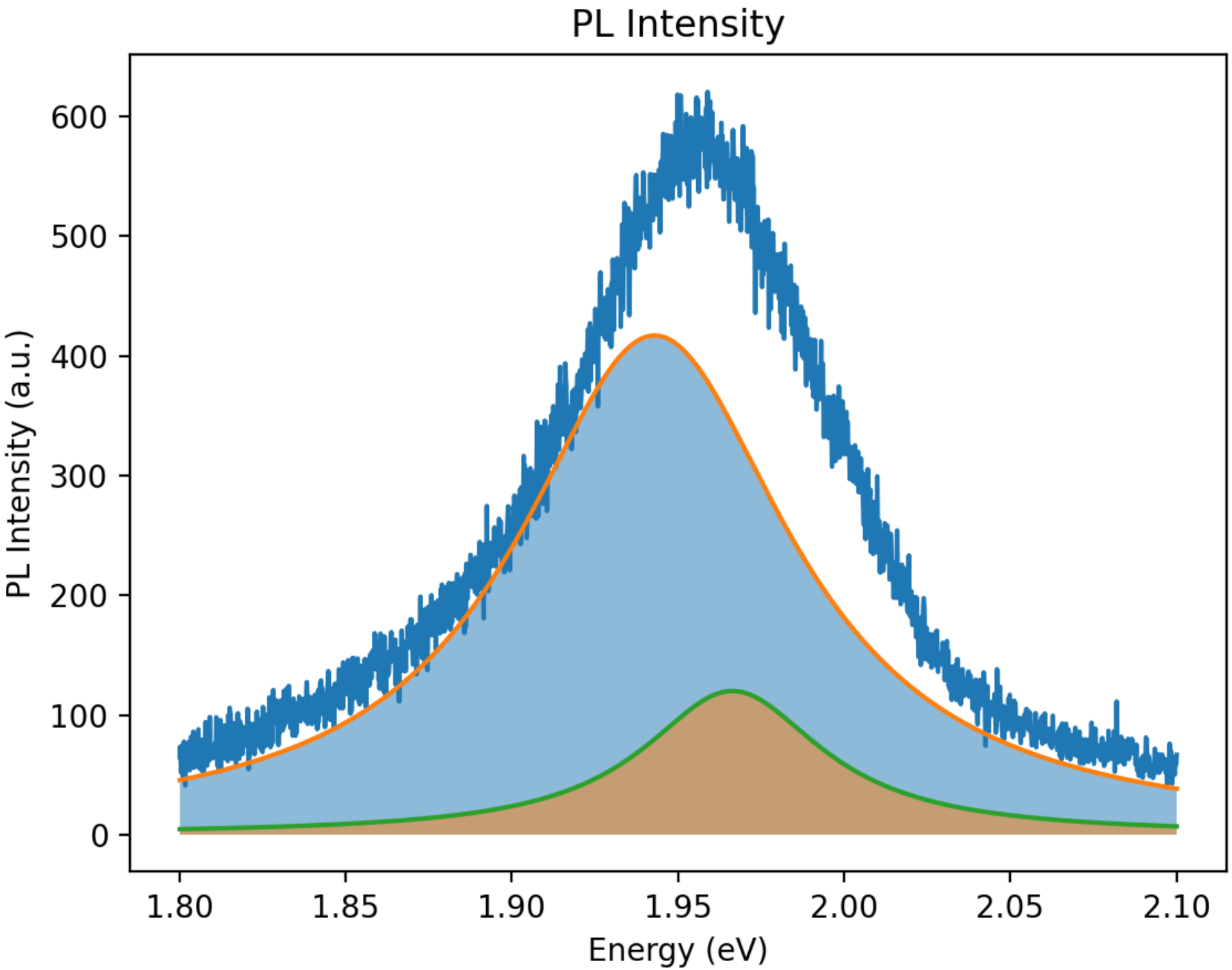

D
X_S^-
X_T^-
X_0
Re-doped
Pristine
Normalised PL intensity (a.u.)
400
300
200
100
0
-100
Photon energy (meV)

Re-WS2 PL fit (Spot1, Laser power 20mW, 0.5%, RT)



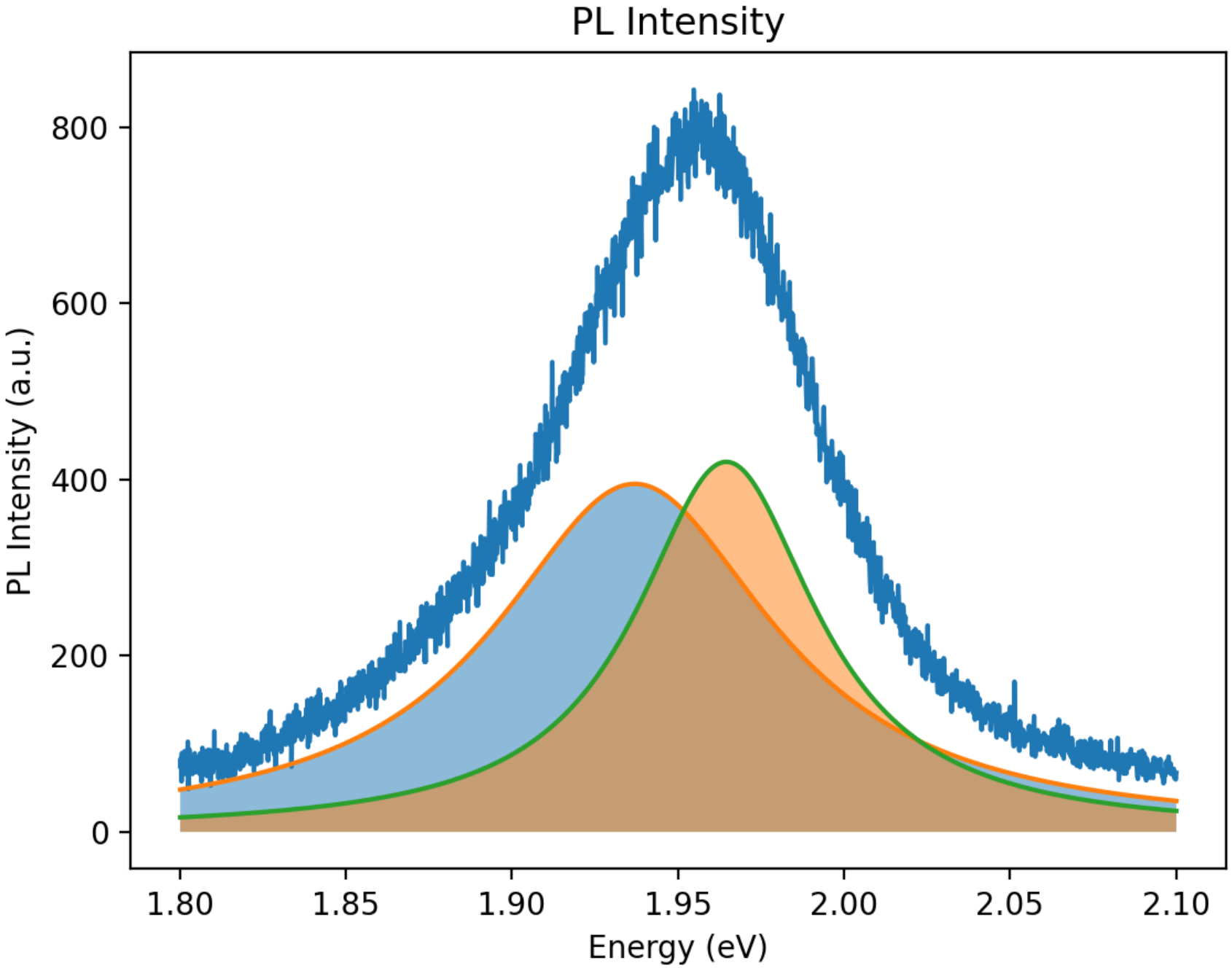
1.930eV: Linewidth: 106meV
1.958eV: Linewidth: 71meV

Re-WS2 PL fit (Spot1, Laser power 20mW, 0.1%, RT)



1.943eV: Linewidth: 103meV
1.966eV: Linewidth: 66meV

Re-WS2 PL fit (Spot2, Laser power 20mW, 0.1%, RT)

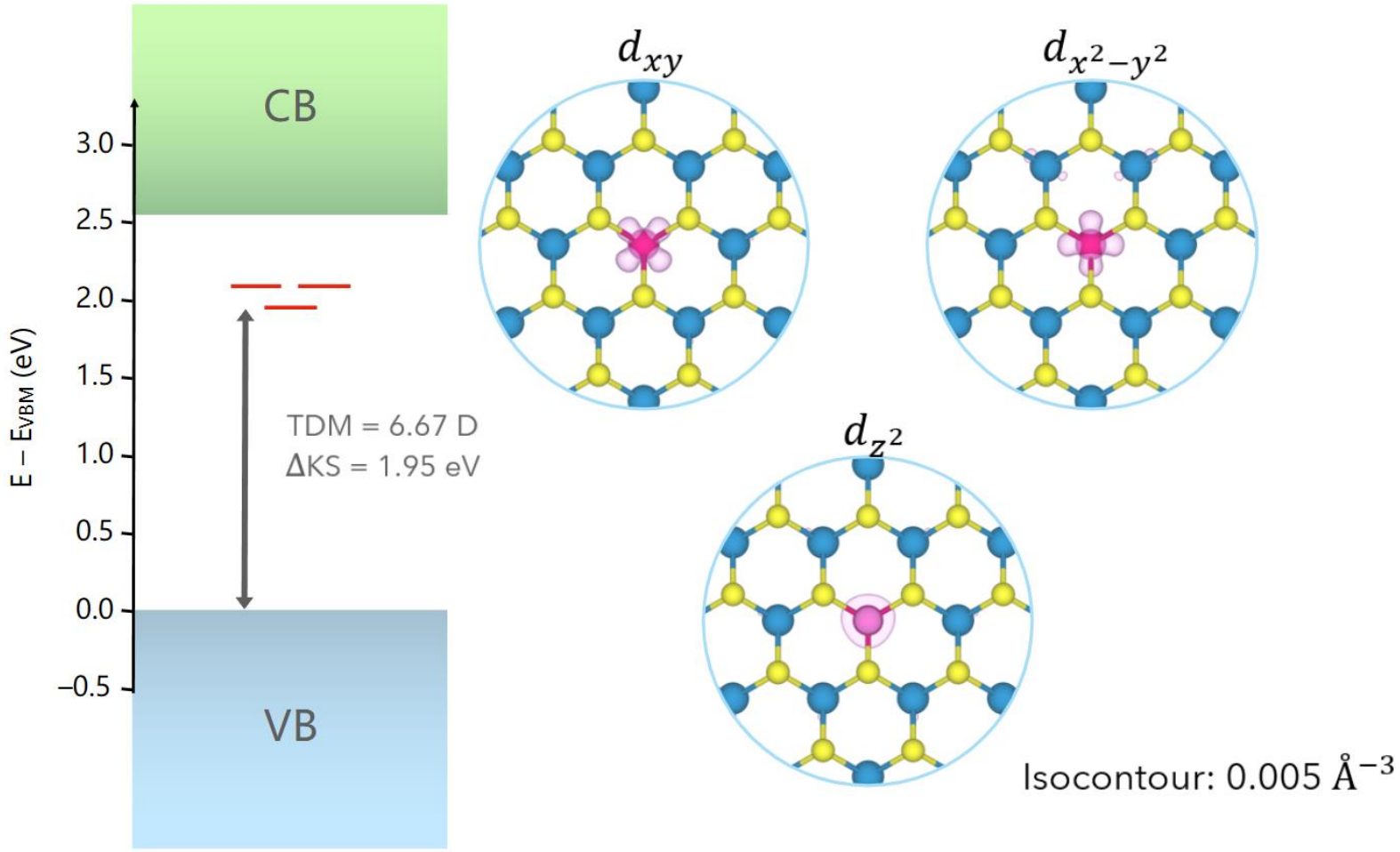


1.941eV: Linewidth: 102meV
1.965eV: Linewidth: 66meV

Calculation

Ground state electronic structure in WS_2

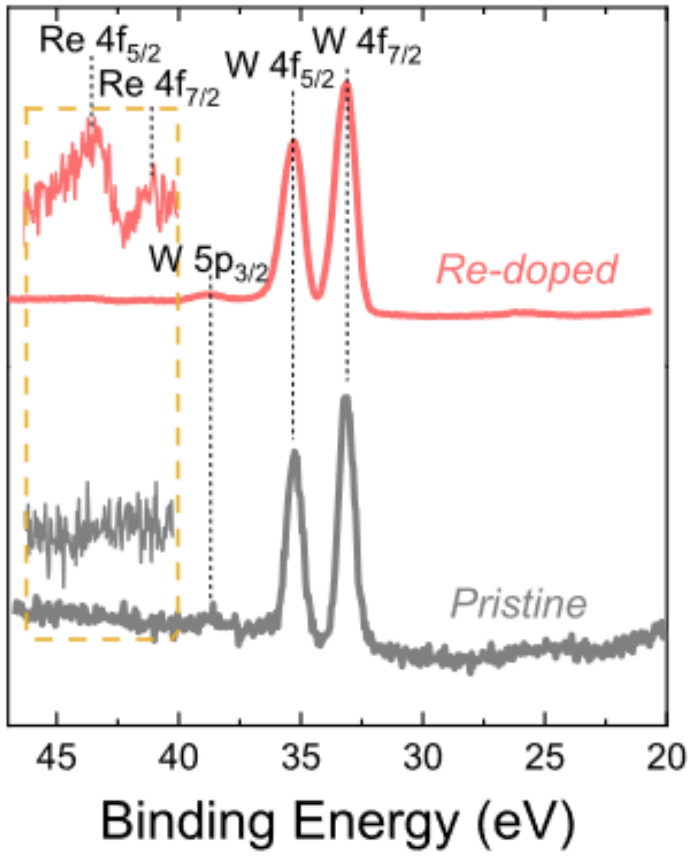
Re_W^+ in WS_2 has a singlet ground state with three in-gap defect states originating from Re_d orbitals.

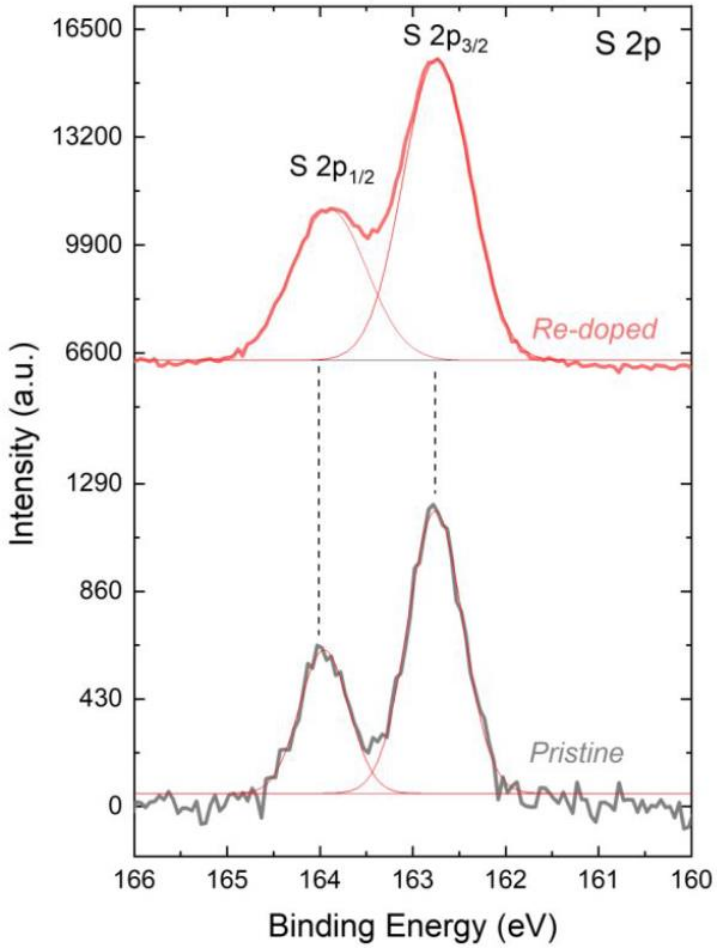


Impurity-Induced Emission in Re-Doped WS_2 Monolayers

Leyi Loh, Yifeng Chen, Junyong Wang, Xinmao Yin, Chi Sin Tang, Qi Zhang, Kenji Watanabe, Takashi Taniguchi, Andrew TS Wee, Michel Bosman, Su Ying Quek, and Goki Eda*

Cite This: *Nano Lett.* 2021, 21, 5293–5300 Read Online

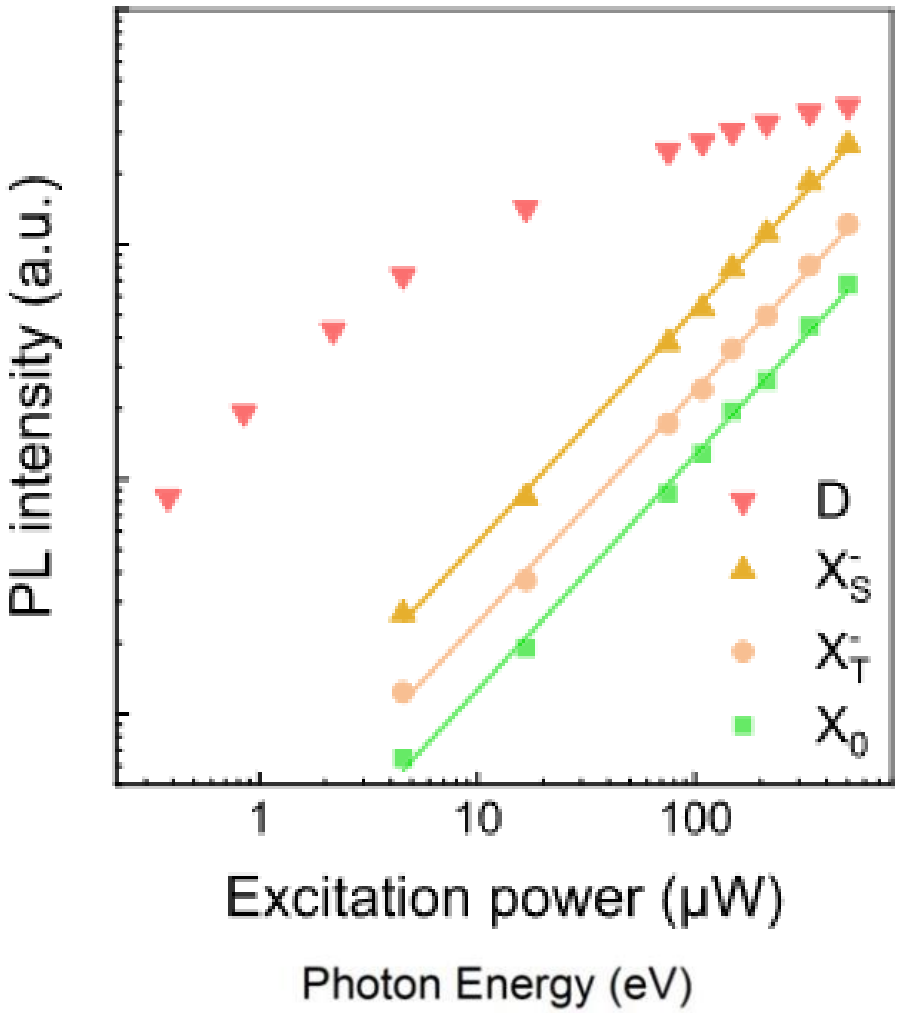




HR-XPS: Spectra shows Re $4f_{5/2}$and $4f_{7/2}$ peaks at 44.1 and 41.2 eV, verifying the successful incorporation of Re in the lattice

Doping level ~1%
Concentration $\sim 1 \times 10^{-13} cm^{-2}$

Monolayer PL (4K)

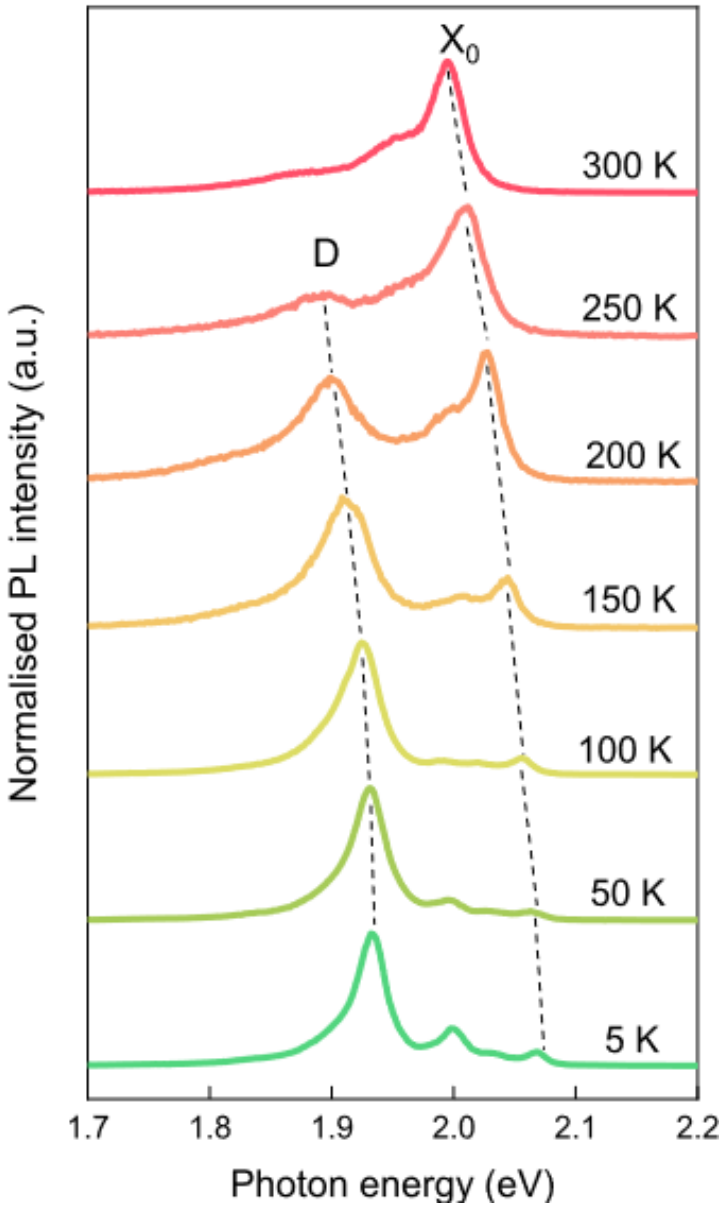


New strong peak (D) below the free exciton (X0) and trion (X_S^-, singlet; X_T^- , triplet) peaks

Peak D is located 140 meV below the free exciton peak.
FWHM=40meV
(40meV is smaller than inhomogeneously broadened defect peaks in unencapsulated samples, although larger than typical quantum emitters in TMDs)

Intensity of D peak shows a sublinear increase with the excitation power, which is a common characteristic of localized states

TR-PL & Temperature-dependent

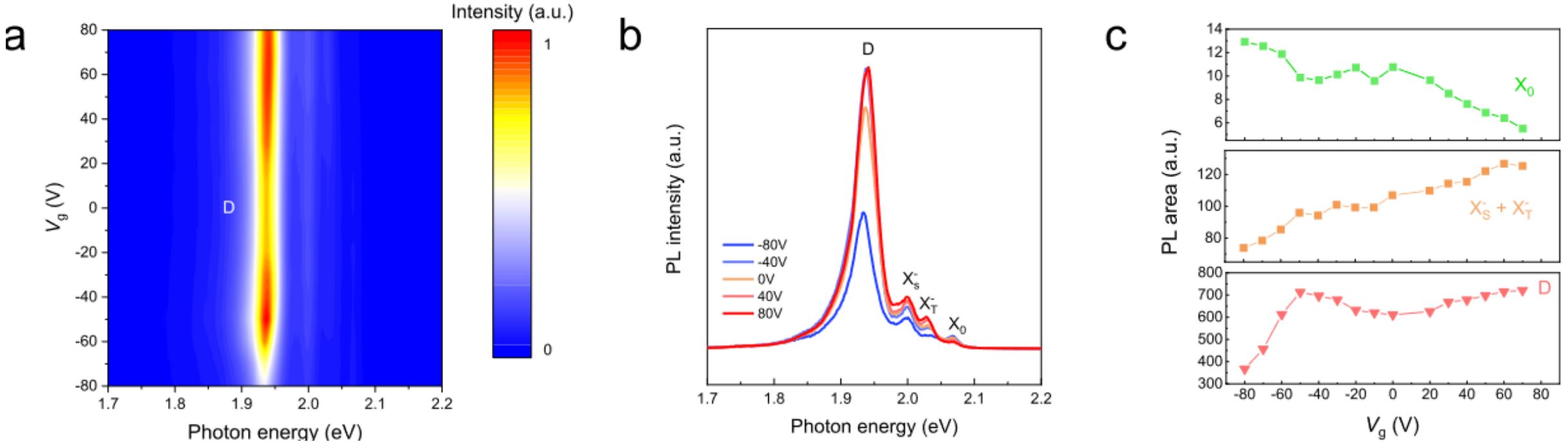


The average emission lifetime of the D peak is 1.53ns, longer than that of the singlet trion peak (247 ps)

The decay curve of D was fitted with a triexponential function with fast (τ_1 = 327 ps), intermediate (τ_2 = 1.54 ns), and slow (τ_3 = 7.48 ns) time constants

Temperature↑ → Intensity↓
T >250K spectra dominated by X_0(free exciton)

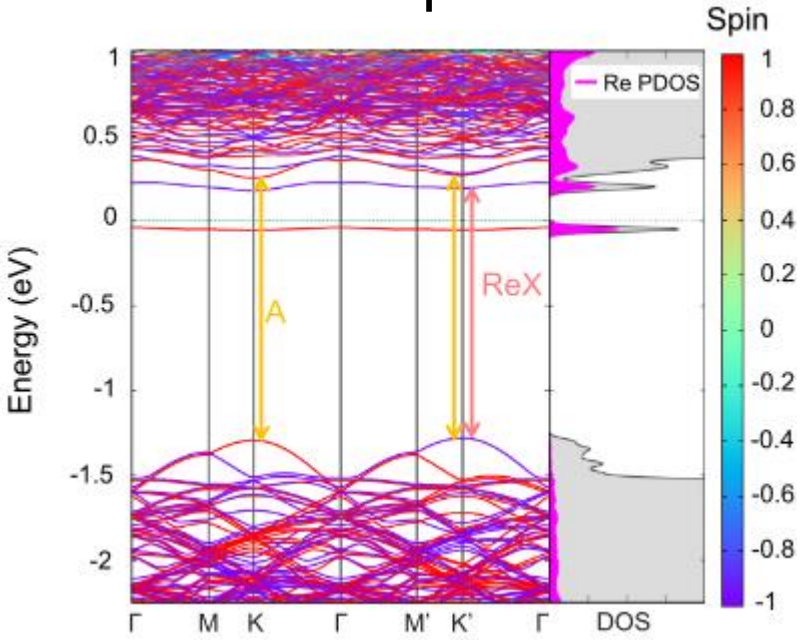
Gate-dependent emission spectrum

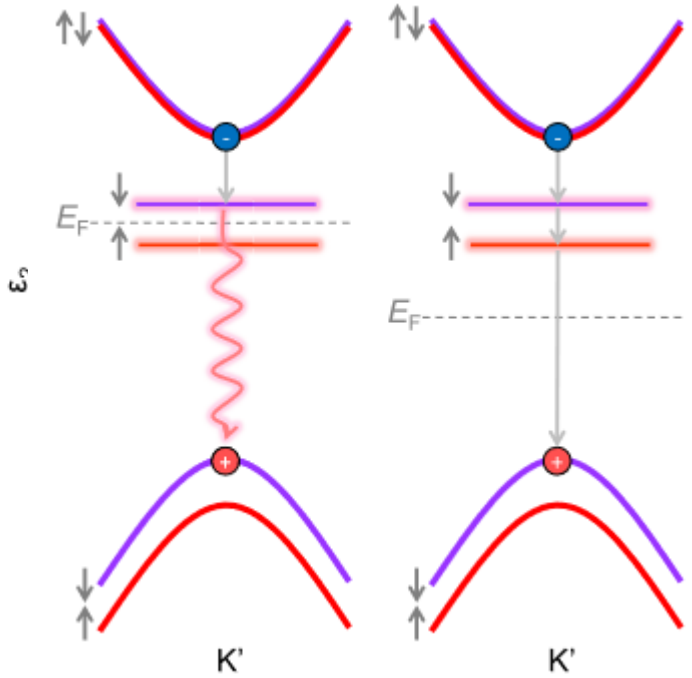


The D peak remains nearly constant but quickly quenches below $V_g \sim -60$ V. (attributed to the electrostatic ionization of the donor state)

The behavior of the D peak, which appears to be uncorrelated with the changes in the X_0 and $X_S^- + X_T^-$ peak intensities, suggests that its emission mechanism involves a neutral donor state.

First Principles Calculations





Re defect-related exciton (ReX) peaks ~0.15 − 0.32 eV below A

These ReX peaks involve transitions from VBM to the unoccupied defect band in the neighborhood of K′, while the A peak stems from transitions at both the K and K′ points

The occupied Re band is located 300 meV below the CBM, indicating a high activation energy for the ionization of the Re donor

When no gate voltage is applied, the Fermi level (E_F)is between the spin-split Re states, and the electron from the CBM can relax to the upper unoccupied Re state and recombine with a hole at the VBM upon optical pumping. **(Left)**

When the Fermi level is tuned to the midgap position and Re is positively charged, the electron in the upper Re state can further relax to the lower empty Re state**.(Right)**

Their unsolved questions

TMDC	Energy (meV)		Binding energy (meV)					
	E_{D^0}	E_{A^0}	E^b_{D+X}	E^b_{A-X}	$E^b_{D^0X}$	$E^b_{A^0X}$	E^b_{D+XX}	E^b_{A-XX}
MoS_2	−638.5	−670.1	7.2	2.7	32.4	31.7	51.8	48.0
$MoSe_2$	−636.8	−659.3	6.5	3.2	31.5	31.1	50.8	48.1
$MoTe_2$	−447.8	−443.9	3.9	4.6	17.9	18.0	32.4	32.9
WS_2	−606.8	−633.6	6.2	2.7	32.0	31.5	48.6	45.3
WSe_2	−542.3	−563.6	5.5	2.6	27.5	27.1	43.4	40.8
WTe_2	−519.0	−562.1	7.9	1.6	26.1	25.3	44.0	38.8

High energy difference(140meV)

Simulations predicted a 30 meV red-shift of such excitons from the A exciton peak due to weak dipole−dipole interactions

Relatively high thermal stability(250K)

An alternative explanation is a transition that involved ionized donor-bound excitons. And these bound excitons should have an explicit gate-voltage dependence for PL intensity, which was not observed in their samples.

PhysRevB.96.075431(2017)

Methods

Substitutional Doping of Re at the W site. Re-doped WS_2 monolayers were synthesized by chemical vapor deposition (CVD), where S vapor was reacted with a dried metal salt precursors on a SiO_2/Si substrate with H_2/Ar (5%/95%) carrier gas at atmospheric pressure and a growth temperature of 850 °C. Similar to our previous work,[30] metal precursor salts, sodium tungstate ($NaWO_4$) and sodium perrhenate ($NaReO_4$) as W and Re cation sources, respectively, were dissolved in deionized water to form a homogeneous clear solution. The solution was then drop-cast onto an UV-ozone-treated SiO_2/Si growth substrate and blow-dried with a N_2 gun to form a thin film of the precursor mixture. The growth consistently yielded truncated triangular-shaped monolayers that were 6−10 μm in size (Figures 1a and S1), similar to recently reported Re-doped WSe_2 monolayers.[31] Reference pristine monolayer samples were prepared by chemical vapor transport (CVT) followed by mechanical exfoliation. The CVT sample was used as a reference because it was more similar to Re-doped WS_2 in its sulfur vacancy concentration than the undoped CVD sample (Figures S2 and S3).[32]

Device Fabrication and Testing. The samples were first coated with a thin film of 950 A5 PMMA and baked at 130 °C for 2 min, followed by the etching of the growth substrate in a hot base (KOH) solution bath to lift the encapsulated (Re:)WS_2/PMMA films. The encapsulated (Re:)WS_2/PMMA film was washed twice, transferred to the desired SiO_2/Si substrate, and subsequently baked for 3 min at 100 °C. To remove the PMMA, the wafer was immersed in acetone for 30 min and then blow-dried by N_2 gas. The monolayer flake was encapsulated by hBN together with two FLG flakes. Cr and Au electrodes with a thickness of 2 and 50 nm, respectively, were deposited on the FLG flakes using standard electron beam lithography and a thermal evaporation process. The device performance was measured in a N_2-filled glovebox at room temperature. The PL was measured in a confocal microspectroscopy setup (NT-MDT) equipped with an Andor iDus CCD camera. The excitation wavelength was 532 nm, and the power was lower than 15 μW. Devices were loaded in a vacuum chamber cooled by liquid helium. Gate and source-drain bias voltages were applied by Agilent B2902 Source/Measure unit. The lifetime measurement was also performed under vacuum at 77 K using a 405 nm picosecond pulse laser (Picoquant). A time-correlated single-photon counting module (PicoHarp 300) and an avalanche photo diode were used as the detection system. For PLE, a frequency-tunable Fianium supercontinuum laser (NKT Photonics) was used as the excitation source with a fixed laser power of 5 μW. The laser line was filtered from the collection path with two 633 nm long pass filters.